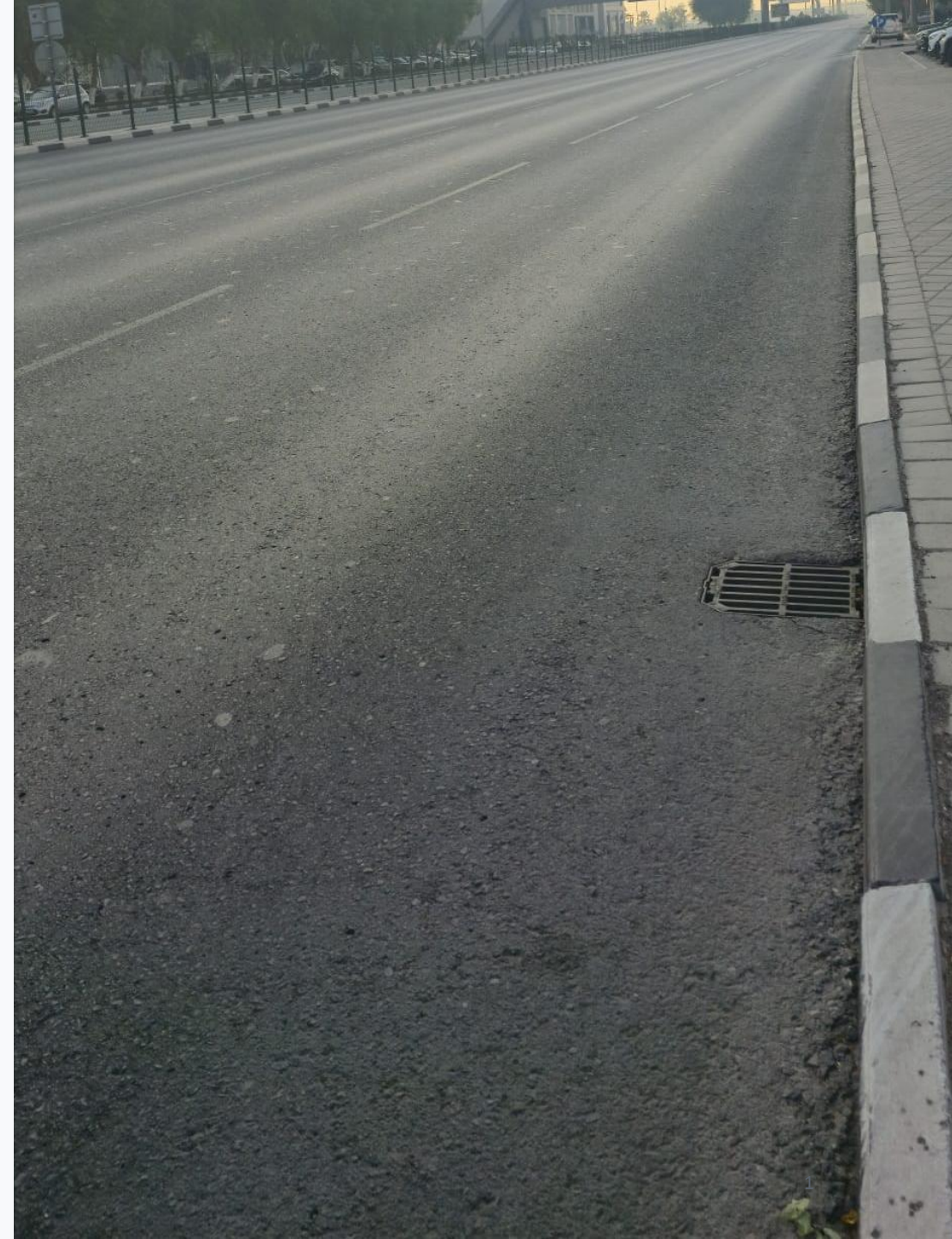


Road Pavement Defect Detection Using YOLOv8

Computer Vision for Preliminary Road-Condition Screening

Pothole • Road Crack • Uneven Manhole



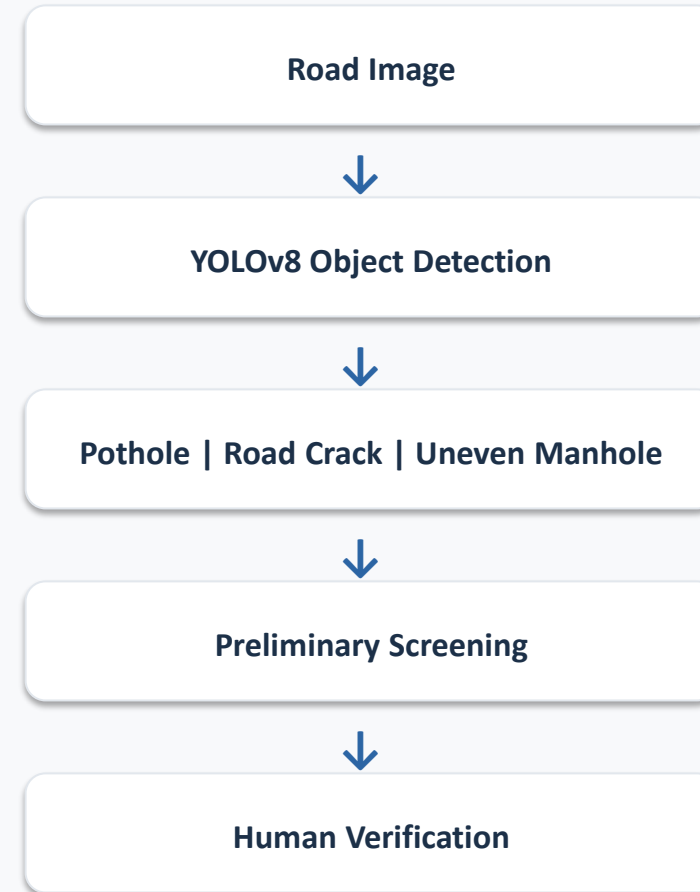
Problem & Computer Vision Approach

Why this problem?

- Road pavement defects can affect road safety, ride quality and maintenance requirements.
- Visual inspection across large road networks can be time-consuming.
- Computer Vision can support preliminary screening by identifying visible defects from road images.

Project objective

Develop and evaluate a YOLOv8 prototype for detecting three visible road-defect classes and identifying locations that may require further inspection.



Dataset Preparation & Annotation



Image sources

- Two public Kaggle datasets
- Additional project road photographs
- 3 classes: pothole, road_crack, uneven_manhole

Five separate project photographs were reserved for unseen inference.

Preprocessing & augmentation

- Auto-Orient
- Resize: 640×640
- Horizontal flip
- Brightness: $\pm 15\%$
- Blur: up to 1.9 px

SAM: useful for some clear boundaries, but manual correction was needed—especially for thin road cracks.

Final exported dataset

606 Training images

50 Validation images

656 Total exported images

252 = original source images; 656 = exported dataset after augmentation.

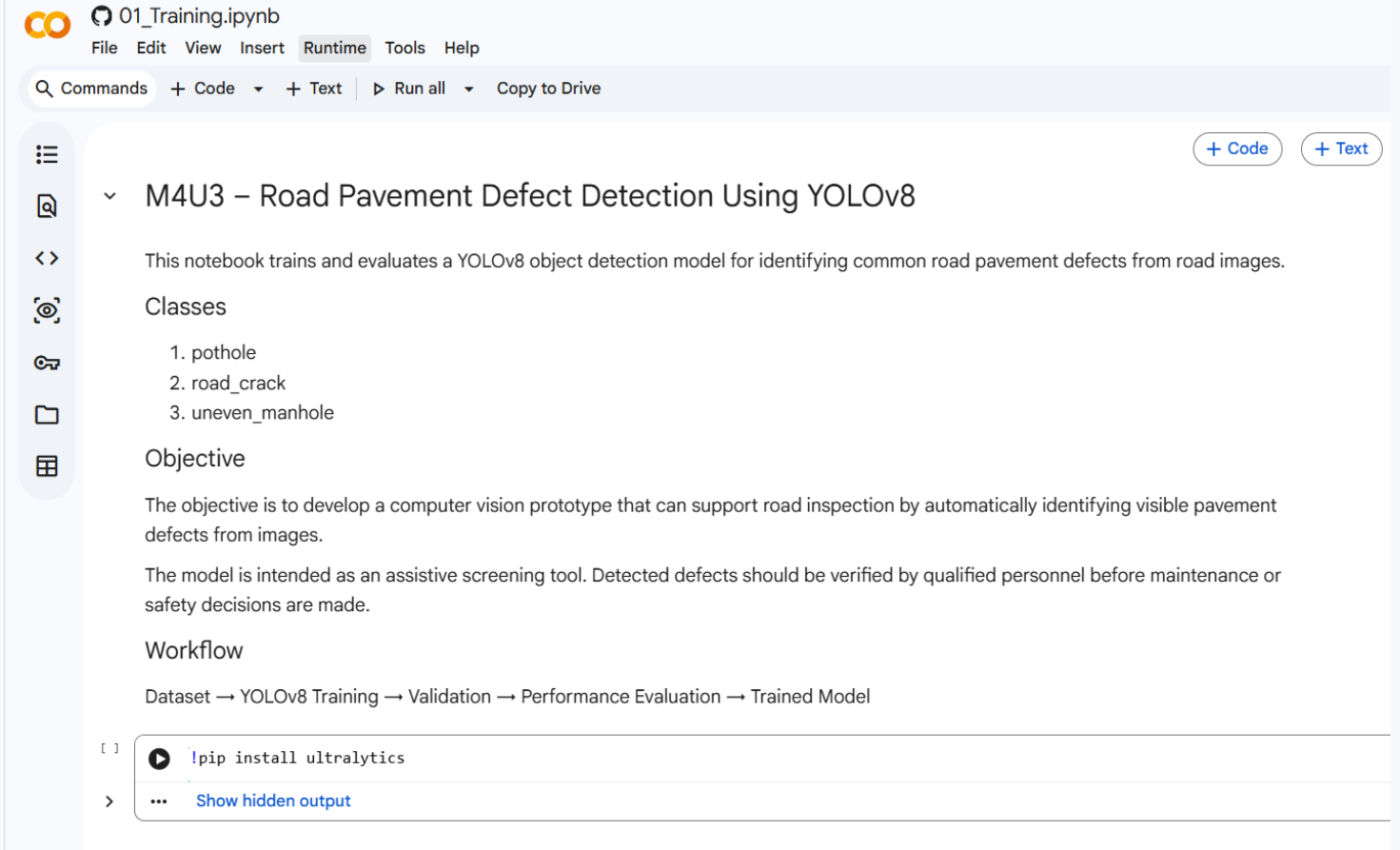
YOLOv8 Model Training

Training setup

Model	YOLOv8 Nano
Epochs	30
Batch size	16
Image size	640 × 640
Environment	Google Colab
GPU	NVIDIA Tesla T4

Workflow

Dataset → Training → Validation →
Performance Evaluation → best.pt



The screenshot shows a Google Colab notebook interface. At the top, the title is '01_Training.ipynb'. Below the title bar, there are tabs for 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. A search bar with 'Commands' is visible. On the left sidebar, there are icons for file management and a table view. The main content area has a title 'M4U3 – Road Pavement Defect Detection Using YOLOv8'. Below the title, there is a description: 'This notebook trains and evaluates a YOLOv8 object detection model for identifying common road pavement defects from road images.' followed by 'Classes' and a list: '1. pothole', '2. road_crack', '3. uneven_manhole'. Then 'Objective' is described: 'The objective is to develop a computer vision prototype that can support road inspection by automatically identifying visible pavement defects from images. The model is intended as an assistive screening tool. Detected defects should be verified by qualified personnel before maintenance or safety decisions are made.' followed by 'Workflow' and a sequence: 'Dataset → YOLOv8 Training → Validation → Performance Evaluation → Trained Model'. At the bottom, there is a code cell with the command '!pip install ultralytics' and a 'Show hidden output' link.

Reproducibility Training notebook downloads the frozen dataset from GitHub Release and runs in Colab without local files.

Model Performance & Validation Results

0.511 Precision

0.434 Recall

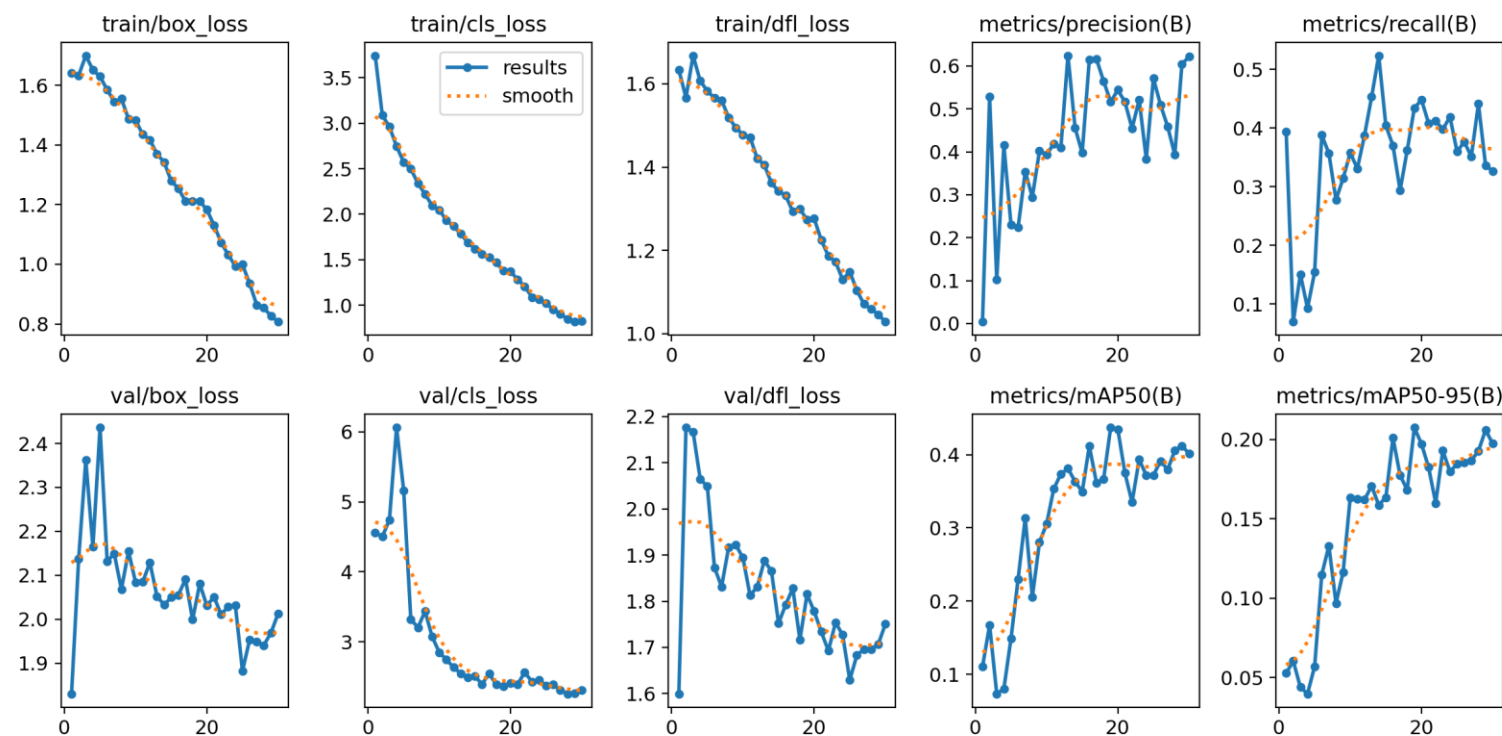
0.438 mAP@50

0.207 mAP@50-95

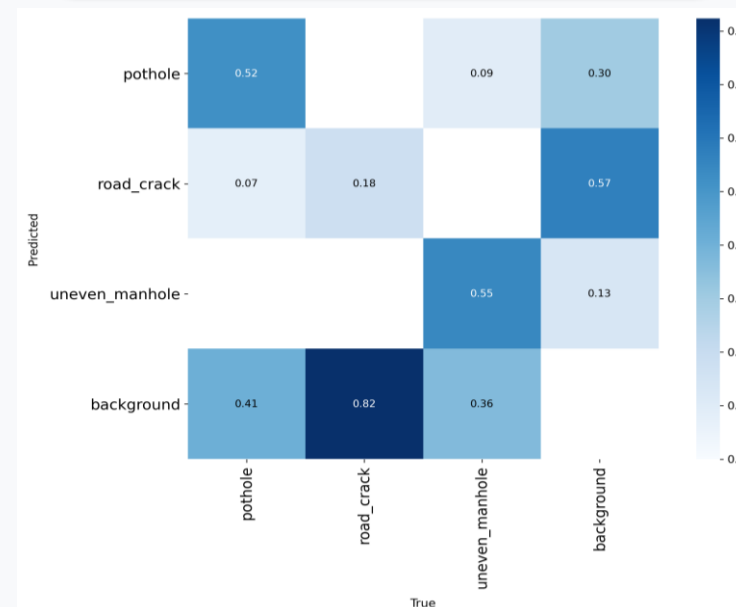
Class-level results

Class	P	R	mAP50
Pothole	.631	.519	.512
Road Crack	.231	.147	.105
Uneven Manhole	.673	.636	.696

Uneven manhole performed strongest; road crack was the most challenging class.



Training curves



Normalized confusion matrix

Detection Evidence & Error Analysis

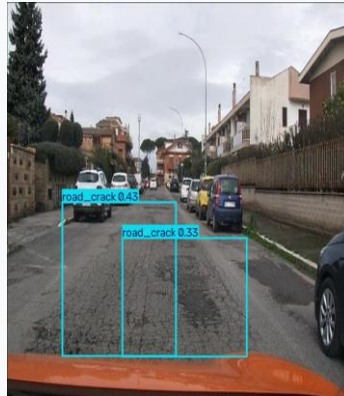
Successful detections



Uneven manhole · 0.89



Pothole · 0.69



Road crack · 0.43 / 0.33

Representative failure cases



Missed road crack
False Negative



Pothole → road crack
Class confusion



Multiple missed
potholes
False Negatives

Key learning

- Road cracks were the main weakness: thin and irregular features were difficult to detect consistently.
- Small and distant defects were sometimes missed in wider road scenes.
- Road joints, repaired surfaces, shadows and markings can visually resemble defects.
- Due to its higher safety risk, an open manhole could be separated as an independent class from uneven_manhole in a future dataset.

Next iteration

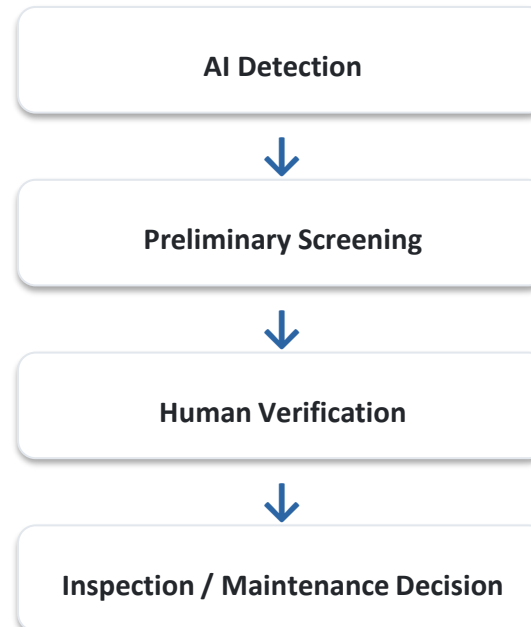
More diverse road-crack images, hard negatives, challenging potholes and consistent annotation.

Limitations, Governance & Next Steps

Current limitations

- Road crack remains the weakest class.
- Small or distant defects may be missed.
- Shadows, markings, joints and repairs may cause visual confusion.
- Mainly trained on daytime road images.
- Not sufficiently tested at night, in rain, on wet pavement, glare or poor visibility.
- 2D images cannot confirm actual manhole level difference.

Human-in-the-loop



Next steps

- More diverse road-crack and challenging defect images.
- Add night, rain, wet-road and varied lighting conditions.
- Standardize annotation format.
- Evaluate further training and model settings.
- Future field use: vehicle-mounted camera + GPS for defect location.

Assistive screening tool only — not the sole verifier for life-safety decisions.

Reproducible in Google Colab: [training + inference notebooks](#), [frozen dataset and weights in GitHub](#).